

Prototype Development of a Food E-Commerce Application for the Elderly Using a User-Centered Design Approach

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Abstract— The use of e-commerce applications in Indonesia has become increasingly widespread across all segments of society, including the elderly, who have specific needs for healthy food to meet their nutritional requirements. Several food e-commerce platforms operate in Indonesia, but their services still do not adequately accommodate the needs of the elderly in certain aspects. Therefore, there is a need to develop more elderly-friendly food e-commerce platforms using a user-centered design approach. The development process began with data gathering to identify the requirements of elderly users, followed by the development of an application based on these requirements, which included designing a low-fidelity prototype and developing a high-fidelity prototype in the frontend. The developed prototype was then tested to gather feedback for evaluation and improvement to ensure the application meets the specified requirements and goals.

Keywords— food e-commerce, elderly, user-centered design, interaction design, prototype

I. INTRODUCTION

Nowadays, the use of e-commerce for daily needs has become increasingly common among the public. Ease of access, a wide selection of products, and time efficiency are several factors that support the use of e-commerce for shopping. E-Commerce use is widespread across all segments of society, including the elderly, who are often considered less adapted to using digital technology. The elderly dominate e-commerce transactions with a total of 13 transactions and 19 product purchases per year on average, the highest compared to other age groups [1].

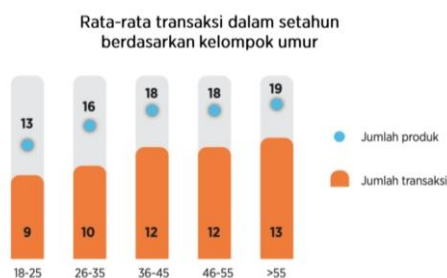


Fig. 1. Average number of transactions and product purchases per year by age group [1]

E-Commerce is important for the elderly to make the transaction process easier than traditional transactions. This is necessary because the physical condition of the elderly is not as good as that of younger people, and their daily needs also differ from those of other age groups, which require more balanced nutrition to support their physical function and mental health, while also reducing the risk of chronic diseases and disabilities [2].

Several food e-commerce platforms already operate in Indonesia, which offer food delivery and supermarket services. However, these existing platforms only accommodate the needs of general users and often fail to consider elderly users, who require simpler ordering processes and interface designs with color schemes that do not hinder visibility.

Based on this problem, there is a need to develop a food e-commerce application to help the elderly order healthy food more easily. The application will be developed using a user-centered design approach that focuses on user needs to provide an interaction suitable for elderly users.

II. EASE OF USE

A. Interaction Design

Interaction design has several meanings based on various information including.

1. Designing interactive products to support the way people communicate and interact in their everyday and working lives.
2. Designing spaces for human communication and interaction.
3. The why as well as the how of our daily interactions using computers.
4. The art of facilitating interactions between humans through products and services.

Interaction design involves various components, including academic disciplines, design practices, and interdisciplinary fields, all of which are related to each other [3].

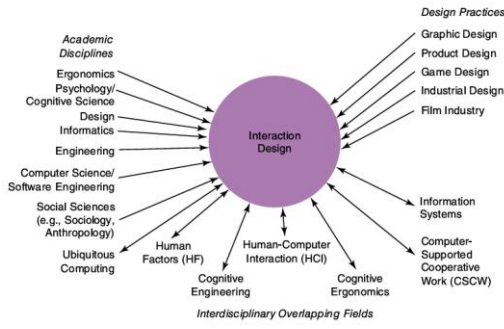


Fig. 2. Relationship among contributing academic disciplines, design practices, and interdisciplinary fields concerned with interaction design [3]

B. User-Centered Design

User-Centered Design (UCD) is an approach that focuses on the needs of users who will be using a product or service. The interaction design is created based on the users' needs, goals, and preferences, which should be identified by the designer. Users are involved in every stage of the product development process to ensure that the product meets their needs [4].

UCD has four basic activities that are related to each other as a lifecycle model [5], which include.

1. Understand context of use
2. Specify user requirements
3. Design solutions
4. Evaluate against requirements

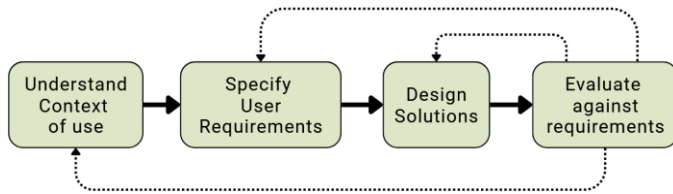


Fig. 3. User-Centered Design lifecycle model [5]

C. Usability and User Experience Goals

Usability Goals are efforts to ensure that an interactive product achieves its intended end goals [3]. Usability goals have seven following goals, including.

1. Effective to use: how good a product is at doing what it is supposed to do.
2. Efficient to use: the way a product supports users in carrying out their tasks.
3. Safe to use: protecting from dangerous conditions and undesirable situations.
4. Having good utility: the extent to which the product provides the right kind of functionality so that users can do what they need or want to do.
5. Easy to learn: how easy a product is to learn to use.
6. Easy to remember how to use: how easy a product is to remember how to use, once learned.

7. Enjoyable to use: how acceptable a product is when being used.

User Experience Goals cover the emotions and felt experiences that users have when interacting with a product, which include desirable and undesirable aspects.

TABLE I. DESIRABLE ASPECTS OF THE USER EXPERIENCE [3]

<i>Desirable Aspects</i>		
<i>Satisfying</i>	<i>Helpful</i>	<i>Fun</i>
<i>Enjoyable</i>	<i>Motivating</i>	<i>Provocative</i>
<i>Engaging</i>	<i>Challenging</i>	<i>Surprising</i>
<i>Pleasurable</i>	<i>Enhancing Sociability</i>	<i>Rewarding</i>
<i>Exciting</i>	<i>Supporting Creativity</i>	<i>Emotionally Fulfilling</i>
<i>Entertaining</i>	<i>Cognitively Stimulating</i>	<i>Experiencing Flow</i>

D. Electronic Commerce (E-Commerce)

Electronic commerce has several meanings based on various information including.

1. Buying and selling of goods and services over the internet [6].
2. Economic activities that find electronic support [7].

E-Commerce, in the context of markets, is seen as a place where supply and demand meet and can be referred to as an “electronic market”, which requires a high level of interactivity settings to quickly connect business activities between sellers and buyers.

The two most commonly used types of e-commerce are marketplace, which facilitates transactions by bringing together many sellers and buyers on a single platform electronically [8], and on-demand service, which directly connects customers with service providers [9].

E. Elderly

Elderly have several meanings based on various information including.

1. Individuals aged 60 years and above who have entered the final stage of human life cycle [10].
2. Decline and weakness in physical condition, with increased vulnerability to diseases and environmental changes [11].
3. Individuals aged 60 years or above, both men and women, who may still be active in daily activities and work, or those who are unable to live independently and must rely on others [12].

Elderly age group can be classified into several categories [13], including.

1. Middle age: 45-54 years
2. Elderly: 55-65 years
3. Young old: 66-74 years
4. Old: 75-90 years
5. Very old: over 90 years

III. ISSUE IDENTIFICATION

Issue identification process involved data gathering using a combination of methods, including.

1. Questionnaire: used to collect quantitative data from a larger sample size [14]. This method was implemented by creating a Google Form, which was filled out by 71 respondents. Data transformation was applied to enhance the accuracy of problem identification.
2. Interview: used to collect in-depth and detailed information [15]. This method involved preparing questions about the use of e-commerce for purchasing daily food, which were asked to 5 respondents during interviews.
3. Observation to collect additional information by examining the current condition of existing applications.

Based on data gathering, the results can be categorized into four sections, including business processes, general activities, behavioral characteristics, and problems that were used to define the solution design.

A. Business Process

TABLE II. BUSINESS PROCESS IDENTIFIED LIST

PB-ID	Business Process
PB-01	Register or login to access a personal account
PB-02	Select a business for purchasing
PB-03	Choose products to order
PB-04	Mark a business as a favorite and check it
PB-05	Confirm the order for the selected products
PB-06	Check the status and location of an existing order
PB-07	Communicate with the driver to obtain information
PB-08	Review of the history of completed orders
PB-09	Provide ratings and comments for previously purchased businesses
PB-10	Check the ratings and comments of a business

B. General Activity

TABLE III. GENERAL ACTIVITY IDENTIFIED LIST

A-ID	General Activity
A-01	Users create a registered personal account that can be used for various activities, such as ordering products and storing data.
A-02	Users can select the business that they want through various methods, such as a recommendation list or a business search menu.
A-03	Users can place product orders by filling in the required order details and selecting a payment method.
A-04	Users can communicate with the delivery driver to obtain information related to their order.
A-05	Users can check detailed information about the delivery status of an ongoing order.
A-06	Users can view the history of their previous orders through the order history menu.
A-07	Users can provide reviews or ratings for businesses they have ordered from previously.

C. Behavioral Characteristic

TABLE IV. BEHAVIORAL CHARACTERISTIC IDENTIFIED LIST

P-ID	Behavioral Characteristic
P-01	Users frequently use e-commerce platforms to shop or purchase products as part of their daily activities.
P-02	Users want an e-commerce platform that offers convenience in the transaction process.

P-ID	Behavioral Characteristic
P-03	Users want clear delivery of information regarding product listings or notification messages so that they are easy to understand.
P-04	Users want an application design that is consistent and aligned with their functional needs.
P-05	Users want flexibility in purchasing product needs, whether it is ready-to-eat meals or groceries that can be prepared at home.

D. Problem

TABLE V. PROBLEM IDENTIFIED LIST

M-ID	Problem
M-01	Difficulty in obtaining detailed information about products being sold, such as images, availability, safety labels, and the address of the business selling the products.
M-02	Issues in the transaction process flow, such as difficulties in customizing orders and completing the payment process.
M-03	Application design issues, particularly in menu layout and color schemes that are unsuitable for elderly users.
M-04	Difficulty in text readability for elderly users due to font size, typeface, and color contrast.
M-05	Unclear delivery information, including driver location, estimated arrival time, and difficulty contacting the driver.
M-06	Security concerns, as the application is not yet fully trustworthy.

IV. SOLUTION DESIGN

The solution design defined here is based on the issues identified in the previous section. It aims to determine the requirements for developing the food e-commerce application, ensuring that it can effectively fulfill user needs.

The solution design defined includes user requirements, system functionality, feature design requirements, as well as usability and user experience goals to be achieved.

A. User Requirements

TABLE VI. USER REQUIREMENTS SOLUTION DESIGN

KP-ID	User Requirements	Traceability
KP-01	Users gain personal security and trust in using the application in an easy way.	PB-01
		A-01
		P-01
		M-06
KP-02	Users can search for businesses and order desired products more easily.	PB-02, PB-03, PB-04
		A-02
		P-01, P-02, P-03, P-04, P-05
		M-01, M-02, M-03, M-04
KP-03	Users can carry out the product ordering transaction process more easily.	PB-05
		A-03
		P-01, P-02, P-03
		M-02
KP-04	Users can access information about the delivery of orders they have previously made.	PB-06, PB-07
		A-04, A-05
		P-03
		M-05
KP-05	Users can receive clear notification information when performing actions, with a display that adapts to their needs.	PB-01, PB-02, PB-03, PB-04, PB-05, PB-06, PB-07, PB-09
		A-03, A-04, A-07
		P-03, P-04
		M-03, M-04, M-05
KP-06	Users can view information related to their previous purchase history.	PB-08
		A-06
		P-03
		M-03, M-04, M-05
KP-07	Users can provide ratings and comments to the driver or business they ordered from previously.	PB-09, PB-10
		A-07
		P-03, P-04
		M-01

B. System Functionality

TABLE VII. SYSTEM FUNCTIONALITY SOLUTION DESIGN

KF-ID	System Functionality	Traceability
KF-01	Registration or login process to access each personal account.	KP-01
KF-02	Logout option to exit the personal account and switch to another account.	KP-01
KF-03	List of recommended businesses.	KP-02
KF-04	Search and sorting for finding businesses.	KP-02
KF-05	List of businesses offering sales promotions.	KP-02
KF-06	Mark or unmark a business as a favorite.	KP-02
KF-07	List of businesses marked as favorites.	KP-02
KF-08	Detailed information about a business with a layout that is easy to view and content that is easy to understand.	KP-02
KF-09	Detailed information about products sold by a business with a layout that is easy to view and content that is easy to understand.	KP-02
KF-10	Customize the product to be ordered.	KP-02
KF-11	Choose the delivery address for an order.	KP-03
KF-12	Add delivery notes for the driver.	KP-03
KF-13	Select a payment method for an order.	KP-03
KF-14	Confirm an order for selected products along with the previously entered information.	KP-03
KF-15	Detailed information about the status and location of a previously placed order.	KP-04
KF-16	Cancel an order that is still in processing.	KP-04
KF-17	Chat with the delivery driver.	KP-04
KF-18	Show a simple and easy-to-understand message upon success or failure of an action.	KP-01, KP-02, KP-03, KP-04, KP-05, KP-07
KF-19	Detailed history of completed orders.	KP-06
KF-20	Reorder from a previously ordered business.	KP-02, KP-06
KF-21	Give rating and comment for the driver or business that ordered from previously.	KP-07
KF-22	View all ratings and comments for a business.	KP-07

C. Feature Design Requirements

TABLE VIII. FEATURE DESIGN REQUIREMENTS

FD-ID	Feature	FD-ID	Feature	FD-ID	Feature
FD-01	Elevated Button	FD-11	Carousel	FD-21	Counter
FD-02	Icon Button	FD-12	Card	FD-22	Dialogs
FD-03	Text Button	FD-13	ListView	FD-23	Snackbar & Toast Message
FD-04	Outlined Button	FD-14	ListTitle	FD-24	Loading Indicator
FD-05	Input Field	FD-15	Business Selection	FD-25	Image
FD-06	Search Bar	FD-16	Sort Option	FD-26	Chat Bubble
FD-07	Notification	FD-17	Bottom Sheet	FD-27	Rating Bar
FD-08	App Bar	FD-18	Chips	FD-28	Order Status Selection
FD-09	Bottom Bar	FD-19	Radiobutton Selection	FD-29	Maps & Pinpoints
FD-10	Bottom Navbar	FD-20	Checkbox Selection		

D. Usability and User Experience Goals

TABLE IX. USABILITY AND USER EXPERIENCE GOALS

Goals		Traceability	
Usability Goals	Effective to use	P-02, P-03	M-01, M-02, M-05
	Efficient to use	P-02	M-02
	Safe to use	-	M-06

Usability Goals	Enjoyable to use	P-03, P-04, P-05	M-01, M-03, M-04, M-05
	Easy to learn	P-01, P-02	M-02
UX Goals	Helpful	P-02, P-03, P-05	M-01, M-02, M-05
	Engaging	P-04	M-03, M-04

V. PROTOTYPE IMPLEMENTATION

A. Low-Fidelity Prototype

Low-Fidelity Prototype development process aimed to provide an overview of interaction design and a detailed application flow. The prototype was created using Figma and implemented for all required pages based on the previously identified system functionalities and feature design requirements. A total of 30 prototype pages were created, as listed in Table X with navigation diagram shown in Fig. 4.

TABLE X. LIST OF PROTOTYPE PAGES

LF-ID	Page Name	LF-ID	Page Name
LF-01	Splash screen and initial login	LF-16	Order confirmation menu page
LF-02	Login page with phone number	LF-17	Delivery address bottom sheet
LF-03	Phone number OTP verification	LF-18	Delivery notes bottom sheet
LF-04	Main menu home page	LF-19	Payment method selection bottom sheet
LF-05	Personal profile	LF-20	Business promotions menu
LF-06	List of favorite businesses page	LF-21	Detailed promotions page for business
LF-07	Details page for recommended businesses	LF-22	Order activity history
LF-08	Search page for businesses	LF-23	Previous order activity details
LF-09	Delivery address bottom sheet	LF-24	Rating and comment submission
LF-10	Search sort bottom sheet	LF-25	Ongoing orders list
LF-11	List of products sold by the business	LF-26	Order transaction details (In Process)
LF-12	List of ratings and comments	LF-27	Order cancellation and confirmation dialog
LF-13	Message for adding or removing favorite business	LF-28	Order transaction details (On the Way)
LF-14	Product order details page	LF-29	Full contacts list
LF-15	Bottom bar for order confirmation	LF-30	Chat details

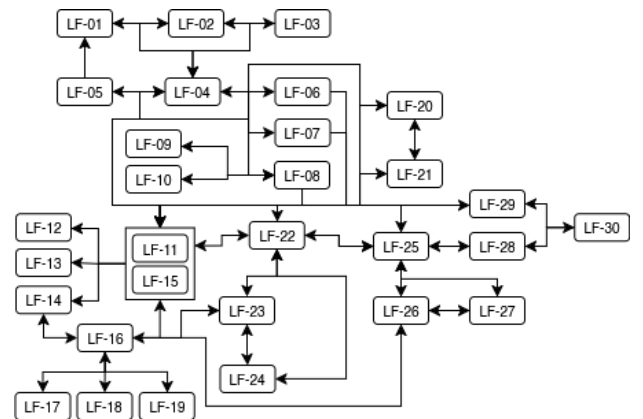


Fig. 4. Navigation diagram for prototype pages

B. High-Fidelity Prototype

High-fidelity prototype development process aimed to represent a realistic user interface with detailed and interactive elements, including visual design, interface components, and user interaction flow. The prototype was implemented as frontend modules to cover all pages from the low-fidelity prototype and was developed using the Flutter framework, which was selected based on development requirements focusing on the Android operating system.

The application developed is named “Nutrisenior”, a combination of the words “nutrition” and “senior”. The name reflects the application’s purpose, which is to fulfill the nutritional needs of the elderly.



Fig. 5. Nutrisenior application logo [16]

The logo represents the elderly in blue, which symbolizes the fulfillment of their nutritional needs, while the green leaf at the base signifies support for providing healthy food through the application.

There are several visual aspects in the application that are important for the user experience, including.

1. Layout and navigation

Prototype layout implemented in the application aims to create an interface that is easy for users to understand by using elements that are proportionally designed and support usability. Several types of layout formats used in the application are explained in Table XI.

TABLE XI. LIST OF PROTOTYPE PAGES

Layout Type	Application Usage
Grid	Displays recommendations and promotions for businesses and products being sold.
List	Shows the complete list in sequential order.
Card	Presents important information such as business details, products, and drivers.
Tabbed	Separates categories for businesses and history.
Form	Provides input fields such as phone numbers, ratings, and reports.

Navigation in the prototype is designed to support ease of use by presenting only the processes relevant to elderly users. The application also includes a bottom navigation bar for accessing important pages, along with clear icons and text labels to ensure better understanding for elderly users.

2. Color

Color is important for supporting the brand and creating an attractive impression to help users feel more comfortable when using the application. Nutrisenior uses a

combination of white and green for several reasons, including.

- Emotional and mental effects of green can provide a sense of calm to users’ feelings and mental state when using the application [17].
- Impression that the products offer healthy food needed by the elderly.
- Simplicity in application visuals that can help elderly.

The color palette used in the application is divided into two types. Below is the color palette information, including the color names and their hex codes [18].

a. Primary Color

Text #20222E	Text #20222E	Text #20222E	Text #20222E	Text #FCFCFC	Text #FCFCFC
Soapstone #FCFCFC	Ecru White #F3F5E2	Beryl Green #DDEECB	Frog Green #A4C639	Drab Green #8D9E4C	Woodland #42553B

Fig. 6. Primary color palette information [18]

Main text colors are #FCFCFC for light and #20222E for dark with consideration that purely light or dark colors can create high contrast, which may cause visual fatigue for elderly, especially under low-light conditions [19].

The color selection has also been tested for contrast values, with all showing good contrast [20], except for Text Foreground (Soapstone) – Background (Drab Green), which has a value of only 3,52 and is therefore used minimally in the application.

List of color combination usage in the application is shown in Table XII.

TABLE XII. PRIMARY COLOR APPLICATION USAGE

Color Combination	Application Usage
Foreground (Dark) – Background (Soapstone)	Main background color for almost all pages.
Text Foreground (Dark) – Background (Ecru White)	Background color for several pages and the bottom sheet.
Text Foreground (Dark) – Background (Beryl Green)	App bar for indicating the page and bottom bar for transaction processes.
Text Foreground (Dark) – Background (Frog Green)	Information on certain types of cards and indicators when an item in a list is selected.
Text Foreground (Soapstone) – Background (Drab Green)	Title information and user-side chat bubbles with bold text to improve readability.
Text Foreground (Soapstone) – Background (Woodland)	Various Elevated Buttons throughout the pages for performing actions when pressed.

b. Secondary Color

Persian Red #D32F2F	Orangy Yellow #FFD700	Amber #FFC107	Blue Dress #007CE0	Mountain Mist #979797	Lavender Pinocchio #E0E0E0
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Fig. 7. Secondary color palette information [18]

TABLE XIII. SECONDARY COLOR APPLICATION USAGE

Color	Application Usage
Persian Red	Warnings or selecting favorites button.
Orangy Yellow	Important confirmation.
Amber	Stars in rating evaluations.
Blue Dress	Minor information buttons, such as resending notifications or editing a profile.
Mountain Mist	Placeholder or state when a navigation bar button is unselected.
Lavender Pinocchio	Color for when a button is disabled

3. Typography

Typography used in the application is a Sans-Serif type with the Roboto font. It was chosen because it offers better readability for the elderly than other typography styles. Other important aspects of the text in the application are text size and bold weight type, with usage details provided in Tables XIV and XV.

TABLE XIV. TEXT SIZE APPLICATION USAGE

Text Size	Application Usage
Main Title (40dp – 32px)	Titles on the initial pages for login.
Heading 1 (28dp – 21px)	Username on the main home page.
Heading 2 (24dp – 18 px)	App bar titles on all pages.
Heading 3 (22dp – 17px)	Main titles on several pages with longer text descriptions for better adjustment.
Heading 4 (20dp – 15px)	Title information on the bottom sheet.
Heading 5 (18dp – 14px)	Subtitles for each section on a page.
Paragraph 1 (16dp – 12px)	Important detailed information.
Paragraph 2 (14dp – 11px)	Supporting detailed information.
Paragraph 3 (13dp – 10px)	Less important supporting information.

TABLE XV. BOLD WEIGHT APPLICATION USAGE

Bold Weight	Application Usage
Bold (w700)	Main text weight used for most texts.
Medium (w500)	Medium emphasis that is not very important.
Regular (w400)	Minor descriptions.

4. Iconography

Iconography used in the application is designed to help users understand the features and enhance visual appearance. The icons are sourced from the Icons class in Flutter and the FontAwesomeIcons library, which provides various types of icons that can be utilized as needed.

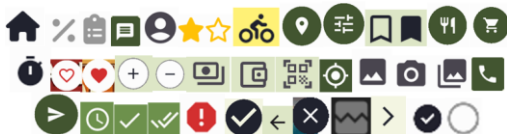


Fig. 8. List of iconographies used in the application

Many icons in the application use solid styles to improve visibility, which are more effective than outline styles in representing function and meaning.

5. Shape Button

Buttons in the application were designed with shapes and sizes suitable for elderly users, featuring large dimensions, proportional spacing, clear color contrast between text labels and the background, and rounded

corners. Applications also provide visual feedback effects, such as color and shadow changes when pressed, to ensure responsive interaction when performing a function, as illustrated in Fig. 9.



Fig. 1. Normal and clicked button states

Some examples of the high-fidelity prototype page designs are shown in Fig. 10.

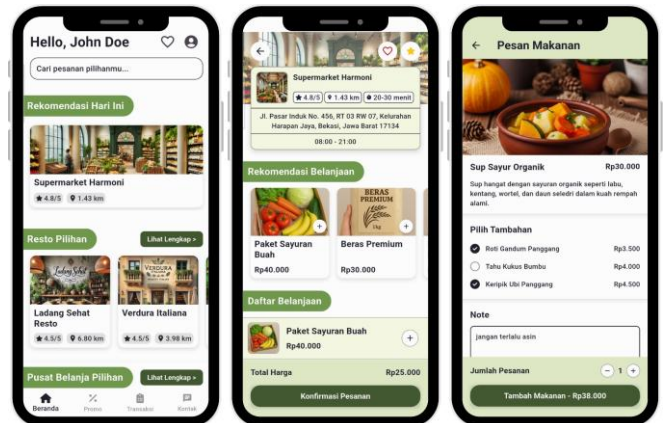


Fig. 10. Some examples of the high-fidelity prototype page

VI. EVALUATION

A. Low-Fidelity Prototype

Evaluation for low-fidelity prototype has a purpose to provide an overview of the interaction design and flow of the developed prototype. It focuses on assessing the suitability and clarity of the application flow to gather feedback for improvements.

Evaluation process will involve elderly participants, with a total of five respondents [21]. They will directly use the prototype to complete specific tasks and provide feedback on the application by answering post-test questions related to ease of use, flow, and any obstacles encountered during the process, with the goal of identifying areas for improvement.

The results of the evaluation show that the elderly can easily understand the overall application flow, and it already helps make purchasing more convenient. Additionally, users suggested adding a reporting feature for businesses, which has been implemented in LF-31: Report Page and included as an additional navigation flow in Fig. 11.

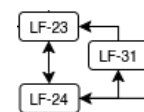


Fig. 11. Additional navigation flow for LF-31: report page

B. High-Fidelity Prototype

Evaluation of the high-fidelity prototype will be conducted through usability testing to obtain user feedback on the quality, comfort, and relevance of the developed interface design to user needs, in order to achieve the defined usability and UX goals. The evaluation uses several metrics based on these goals, as shown in Table XVI.

TABLE XVI. USABILITY AND UX GOALS EVALUATION METRICS

	Goals	Metrics
Usability Goals	Effective to use	Task Completion Rate
	Efficient to use	Time on Task
	Safe to use	Number of Errors
	Enjoyable to use	Customer Satisfaction Score (CSAT)
	Easy to learn	Single Ease Questions (SEQ)
User Experience Goals	Helpful	Intrinsic Motivation Inventory (IMI: value/usefulness)
	Engaging	System Usability Scale (SUS)

The evaluation process will involve five elderly participants [21], who will use the application to complete all defined tasks. Several measurable aspects are recorded during the process. After completing all tasks, participants will also fill out a post-test questionnaire and answer post-test questions. The full list of tasks is shown in Table XVII.

TABLE XVII. LIST OF TASKS

T-ID	Tasks
T-01	Login into personal account.
T-02	Select a recommended business
T-03	Select a business offering promotions.
T-04	Select a specific business using the search menu.
T-05	Select delivery address and sorting conditions.
T-06	Select products from the chosen business.
T-07	Place an order for the products previously selected.
T-08	Check the detailed status of the recently placed order.
T-09	Cancel an order that has already been placed.
T-10	Chat with the driver delivering the order.
T-11	Check an order that has been complete.
T-12	Provide a rating and comment.
T-13	Check the rating and comment previously given.
T-14	Check all ratings and comments for a business.
T-15	Mark a business as a favorite and view the favorites list.
T-16	Report a driver or business that provided unsatisfactory service.

Evaluation was conducted in two iterations, with all results explained based on each evaluation metric.

1. Effective to use

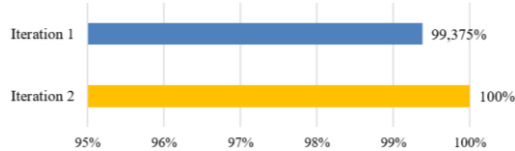


Fig. 12. Comparison of Task Completion Rate

Based on the results, respondents completed all tasks, showing a 0,629% increase in second iteration due to a minor error in task 10 during first iteration when sending multiple messages. This indicates the application is already effective for elderly users.

2. Efficient to use

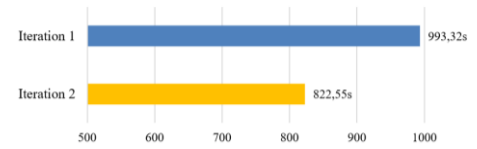


Fig. 13. Comparison of Time on Task

Based on the results, it shows 17,19% decreasing of duration time to done the task faster. It can be happen because affected by some fixing that made in second iteration. This indicates the application is already efficient for elderly users.

3. Safe to use

TABLE XVIII. COMPARISON OF NUMBER OF ERRORS

	Iteration 1	Iteration 2
Number of Errors	1	0

Based on the results, there was a decrease in the number of errors from one in first iteration, which caused by an issue in Task 10 when sending multiple messages, to zero in second iteration. This indicates the application is already safe for elderly users.

4. Enjoyable to use

TABLE XIX. COMPARISON OF CSAT

	Iteration 1	Iteration 2
Customer Satisfaction Score	100%	100%

Based on the results, all five elderly respondents agreed that the application is satisfying and enjoyable to use.

5. Easy to learn

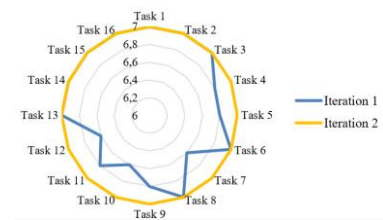


Fig. 14. Comparison of Single Ease Questions (SEQ)

Based on the results, there was a increase in the overall average rating by 1,82%, from 6,875 in the first iteration to 7 in the second iteration. This indicates the application is already easy to learn for elderly users.

6. Helpful

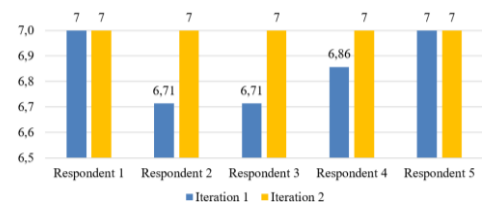


Fig. 15. Comparison of Intrinsic Motivation Inventory (IMI)

Based on the results, there was an increase in the overall average rating by 2,08%, from 85,86 in the first iteration to 87,94 in the second iteration. This indicates the application is already helpful for elderly users.

7. Engaging

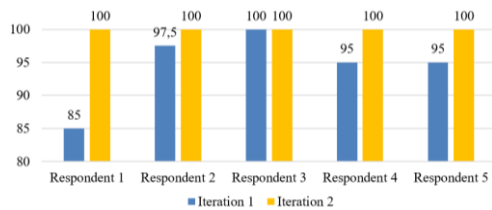


Fig. 16. Comparison of System Usability Scale (SUS)

Based on the results, there was an increase in the overall average rating by 5,82%, from 94,5 in the first iteration to 100 in the second iteration. This indicates the application is already engaging for elderly users.

VII. CONCLUSION

Several conclusions can be drawn from the interaction design process. The main issue in using food e-commerce applications for the elderly concerns the design and application flow. Based on this problem, there is a need to develop a food e-commerce application that is more suitable for elderly users, featuring a layout and navigation optimized for visibility and ease of use, synchronized color palette with a white and green combination, Sans-Serif typography with appropriate size and bold weight, solid iconography instead of outlines, and shape buttons with large dimensions, proportional spacing, clear color contrast, rounded corners, and hover effects for visual feedback.

In addition, five usability goals are identified, which are effective to use, efficient to use, safe to use, enjoyable to use, and easy to learn, and two user experience goals are identified, which are helpful and engaging, that are needed for the development of a food e-commerce application for the elderly.

DELIVERABLES

All deliverables in this project can be accessed at the following link.

1. Technical Documentation:
https://drive.google.com/file/d/11Qlss4Sy9Rr5Bxy_UVJZ-1W2w4VSf0EU
2. Prototype Design:
<https://www.figma.com/design/gJuODaLsdT8PQrgCz9ttUP/NutriSenior-App-for-TA?node-id=0-1&t=xJcGBQeaZ231TDM5-1>
3. Frontend Github Repository:
https://github.com/jasonrivalino/ta_nutrisenior_app

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